

CDF measures

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Claim 1. *The collection of finite unions of (possibly unbounded) intervals of the type $(a, b]$ forms an algebra $\mathcal{A}$ of subsets of $\mathbb{R}$*

Claim 2. *Every $A \in \mathcal{A}$ has finitely many connected components*

$F, G : \mathbb{R} \rightarrow \mathbb{R}$ denote nondecreasing, right continuous functions

Definition 1. *For an interval I with endpoints a and b we define*

$$\text{length}_F(I) = F(b) - F(a)$$

where we use the appropriate limits of F whenever the interval is unbounded

Claim 3. *Define $\mu_F : \mathcal{A} \rightarrow [0, \infty]$ by*

$$\mu_F(A) = \sum_{\text{connected component } I} \text{length}_F(I)$$

Then μ_F is a measure on the algebra $\mathcal{A}$

We now use the Caratheodory extension theorem to extend to $\mathcal{B}(\mathbb{R})$

Claim 4. *Suppose $I = (a, b]$ is bounded. Then we have*

$$\int_I F d\mu_G + \int_I G d\mu_F = F(b)G(b) - F(a)G(a)$$

assuming F and G have no common points of discontinuity

Proof. Let D be the diagonal of the square $I \times I$

- Let S_G denote the set of points of discontinuity of G . A standard result tells us that S_G is countable. Since F and G have no common discontinuities we have $\mu_F(S_G) = 0$
- Now $\mu_D = \int_I \mu_G\{x\} d\mu_F = \int_I 1_{S_G} d\mu_F = 0$
- We have $F(x) = \mu_F(a, x] + F(a)$ and $G(x) = \mu_G(a, x] + G(a)$

Use Tonelli's theorem to rewrite the left hand side as

$$F(a)\text{length}_G(I) + G(a)\text{length}_F(I) + \text{length}_F(I)\text{length}_G(I) + \mu(D)$$

Now simplify the equation to get the desired result

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